Stochastic Volatility Models Based on Martingale Optimal Transport

Alexander Polyakov

Work under supervision of Mikhail V. Zhitlukhin
Laboratory of Stochastic analysis and its applications, HSE

October 12, 2026

Calibration Problem

We consider a market with a risk-free bond and an asset modelled by a semimartingale:

$$B_t = B_0 e^{rt}, \qquad dS_t = \mu_t S_t\, dt + \sigma_t S_t\, dW_t.$$

A **Call option** with strike K and maturity T pays $(S_T - K)^+$. Under a risk-neutral measure the model assigns to it the price

$$C_{\mathrm{m}}(T,K) = e^{-rT}\mathrm{E}[(S_T - K)^+],$$

while the prices $C_{\mathrm{mkt}}(T,K)$ of the same options are quoted in the market.

Calibration problem. Choose the process (S_t) so that its prices reproduce the quoted ones:

$$\sum_{(T,K)} \ell\big(C_{\mathrm{m}}(T,K),\, C_{\mathrm{mkt}}(T,K)\big) \to \min,$$

for a loss function ℓ.

Existing Methods: Local Volatility [B. Dupire 1994]

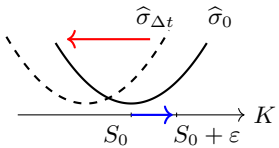
Theorem. Given a surface $C(T, K)$ of class $C^{1,2}$ non-decreasing in T and convex in K, there exists a unique $\sigma_{\text{loc}}(t, s)$ such that

$$dS_t = rS_t\,dt + \sigma_{\text{loc}}(t, S_t)S_t\,dW_t$$

prices all options correctly.

Limitations.

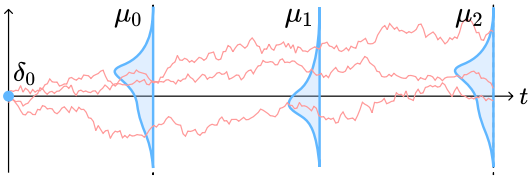
1. Requires a smooth arbitrage-free surface $C(T, K)$: market quotes must be interpolated first.
2. Predicts incorrect smile dynamics: when $S_0 \to S_0 + \varepsilon$, the smile shifts in the opposite direction.



LSV models add a stochastic factor, $\sigma(t, S_t)\,\psi(v_t)$, which repairs the dynamics, but $\sigma(t, S_t)$ still requires a smooth arbitrage-free interpolation of prices.

Optimal transport approach

From now on, $r = 0$: under the pricing measure the asset must be a **martingale**.



Lemma (Breeden–Litzenberger) If $C(T, K)$ is twice differentiable with respect to K, then

$$\partial_{KK} C(T, K) = f_{S_T}(K),$$

where f_{S_T} is the risk-neutral density of S_T.

So the market data are encoded by the marginals $\mu_{T_i} = \mathrm{Law}(S_{T_i})$ at the quoted maturities.

The problem is to find a martingale X such that $X_{T_i} \sim \mu_{T_i}$.

Optimal transport approach

Martingale optimal transport in model calibration

- J. Backhoff-Veraguas, M. Beiglböck, M. Huesmann, S. Källblad. *Martingale Benamou–Brenier: a probabilistic perspective.* Ann. Probab., 2020.
- A. Conze, P. Labordère. *Bass construction with multi-marginals.* SSRN, 2021.
- B. Acciaio, A. Marini, G. Pammer. *Calibration of the Bass local volatility model.* arXiv, 2023.
- M. Beiglböck, B. Jourdain, W. Margheriti, G. Pammer. *A fundamental theorem for weak martingale optimal transport.* Ann. Appl. Probab., 2025.

The calibrated model is selected by its distance to a reference process — this is where the dynamics enter.

Bass Construction [Backhoff-Veraguas et al., 2020; Conze–Labordère, 2021]

Definition Let $\mu_0 \preceq_{cx} \mu_1 \in \mathcal{P}_2(\mathbb{R})$. *Standard stretched Brownian motion* is a process S_t

$$S_t = \mathrm{E}\left[g(Y_1) \mid \mathcal{F}_t\right], \quad \text{where} \quad Y_t = A + B_t \quad \text{and} \quad g = F_{\mu_1}^{-1} \circ F_{Y_1}$$

and the random variable A is taken independent from B_t such that

$$S_0 \sim \mu_0 \quad \text{and} \quad S_1 \sim \mu_1.$$

1. Optimality. The Standard stretched Brownian motion solves the *Martingale Benamou-Brenier problem*:

$$\inf_{\substack{X_t = X_0 + \int_0^t \sigma_s \, dB_s \\ X_0 \sim \mu_0,\ X_1 \sim \mu_1}} \mathrm{E}\left[\int_0^1 (\sigma_t - 1)^2 \, dt\right] = \inf_{X_0 \sim \mu_0,\ X_1 \sim \mu_1} \mathrm{E}\langle X - B \rangle_1.$$

2. Fast calibration. The distribution function of the r.v. A is a fixed point of an operator acting on distribution functions:

$$(\mathcal{A}F)(x) = F_{\mu_0} \circ \Big(\varphi * \big(F_{\mu_1}^{-1} \circ (\varphi * F)\big)\Big)(x),$$

The Bass model still has wrong dynamics of the volatility smiles.

From Bass to SV-Bass

We replace the reference by a **stochastic volatility** model (Y_t, ξ_t):

$$\inf_{X_0 \sim \mu_0,\, X_1 \sim \mu_1} \mathrm{E}\big\langle X - B \big\rangle_1 \qquad \longrightarrow \qquad \inf_{X_0 \sim \mu_0,\, X_1 \sim \mu_1} \mathrm{E}\big\langle X - Y \big\rangle_1,$$

Contributions.

1. Introduce a new cost functional that accounts for the IV smile dynamics.
2. Formulation of a static version of the problem using *Weak optimal transport* and proof of existence of the optimizer.
3. Calibration procedure based on a fixed-point equation, extending the Bass LV scheme to the stochastic volatility setting.

Problem Setup

Given a probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,1]}, \mathrm{P})$ supporting two Brownian motions B, W with correlation ρ. Let $\mu_0 \preceq_{cx} \mu_1$ with $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$.

Define the **reference model** $Y = (Y_t)_{t\in[0,1]}$:

$$\begin{aligned} dY_t &= \xi_t \, dB_t, \\ d\xi_t &= b(\xi_t)\, dt + \eta(\xi_t)\, dW_t, \quad \eta > 0 \end{aligned} \tag{Ref}$$

with initial conditions $Y_0 = 0$, $\xi_0 \sim \beta \in \mathcal{P}_2(\mathbb{R})$.

Define $\mathcal{X}(\mu_0, \mu_1; \kappa)$ as the class of square-integrable martingales $X = (X_t)_{t\in[0,1]}$:

$$X_t = X_0 + \int_0^t \sigma_s^{(1)}\, dB_s + \int_0^t \sigma_s^{(2)}\, dW_s, \qquad \mathrm{E}\int_0^1 (\sigma_t^{(i)})^2\, dt < \infty, \quad i = 1, 2,$$

with $\mathrm{Law}(X_0, \xi_0) = \kappa$ and $\mathrm{Law}(X_1) = \mu_1$, where $\kappa \in \mathcal{P}_2(\mathbb{R}^2)$ has first marginal μ_0.

Dynamic problem. We seek the process in $\mathcal{X}$ that minimizes the angle bracket cost:

$$\inf_{X\in\mathcal{X}(\mu_0,\mu_1;\kappa)} \mathrm{E}\big\langle X - Y \big\rangle_1 \quad \text{equivalently} \quad \sup_{X\in\mathcal{X}(\mu_0,\mu_1;\kappa)} \mathrm{E}\langle X, Y\rangle_1.$$

Static formulation

Weak martingale optimal transport

1. Let $\kappa = \mathrm{Law}(X_0, \xi_0)$ be fixed and define the set of static couplings:

$$\mathrm{Cpl}_{M_1}(\kappa, \mu_1) = \Big\{ \lambda \in \mathrm{Cpl}(\kappa, \mu_1) : \quad \lambda(dx, da, dy) = \lambda_{x,a}(dy)\kappa(dx, da) \\ \int y \, \lambda_{x,a}(dy) = x \quad \kappa\text{-a.e.} \Big\}.$$

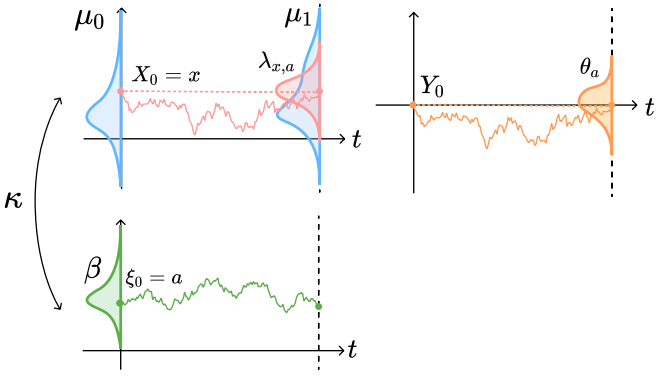
2. For each volatility state a, set $Z^a = \int_0^1 \xi_s^a \, dB_s$ and $\theta_a = \mathrm{Law}(Z^a)$.

Static problem. Transport each conditional law toward the reference increment:

$$\inf_{\lambda \in \mathrm{Cpl}_{M_1}(\kappa, \mu_1)} \iint \mathcal{W}_2^2(\lambda_{x,a}, \theta_a) \, \kappa(dx, da).$$

A *weak optimal transport* problem: the cost depends on the conditional law, not on points. An equivalent maximization form is used in the proofs.

Illustration



Main results

Theorem 1 (Equivalence)**.** The *dynamic* and *static* problems are equivalent, and the optimal process X^* can be constructed from the static optimizer λ^*.

Assumption: $b, \eta : \mathbb{R} \to \mathbb{R}$ are globally Lipschitz, so that for every $\xi_0 = a$ the volatility equation has a unique strong solution ξ^a and transition kernel θ_a is well defined.

$$d\xi_t = b(\xi_t)\, dt + \eta(\xi_t)\, dW_t$$

Theorem 2 (Existence)**.** Let $\mu_0 \preceq_{cx} \mu_1 \in \mathcal{P}_2(\mathbb{R})$ and $\kappa \in \mathrm{Cpl}(\mu_0, \beta)$. Then the static problem admits an optimizer $\lambda^* \in \mathrm{Cpl}_{M_1}(\kappa, \mu_1)$.

Once the static optimizer λ^* is found: in one dimension the transport map with $(f_{x,a})_{\#}\theta_a = \lambda^*_{x,a}$ is the monotone rearrangement $f_{x,a} = F^{-1}_{\lambda^*_{x,a}} \circ F_{\theta_a}$, and

$$X_t = \mathrm{E}\big[f_{X_0,\xi_0}(Y_1) \mid \mathcal{F}_t\big] \qquad \text{is admissible and optimal.}$$

Main results

Theorem 3 Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$, and let $\kappa \in \mathrm{Cpl}(\mu_0, \beta)$. Define the reference process:

$$\tilde{Y}_t = r(X_0, \xi_0) + \int_0^t \xi_s \, dB_s, \qquad 0 \le t \le 1.$$

$$\text{Set} \quad \nu = \mathrm{Law}(\tilde{Y}_1), \qquad g = F_{\mu_1}^{-1} \circ F_\nu.$$

Suppose that the function r is chosen so that

$$\mathrm{E}\left[g\left(r(x,a) + \int_0^1 \xi_s^a \, dB_s \right) \right] = x, \qquad \kappa\text{-a.e. } (x,a).$$

Then

$$X_t = \mathrm{E}[g(\tilde{Y}_1) \mid \mathcal{F}_t], \qquad 0 \le t \le 1.$$

is an optimizer of problem

$$\inf_{X \in \mathcal{X}(\mu_0, \mu_1; \kappa)} \mathrm{E}\big\langle X - Y \big\rangle_1.$$

Calibration Algorithm

For a non-random initial volatility $\xi_0 = a$ the algorithm of the Bass model can be applied.

View $r(x, a)$ as the transport from μ_0 to some measure α. Set

$$(K^{\pm}\psi)(x) = \mathrm{E}[\psi(x \pm Z)], \quad Z = \int_0^1 \xi_t^a \, dB_t.$$

Proposition The law $\alpha = \mathrm{Law}(r(X_0, a))$ can be computed through the fixed-point operator:

$$(\mathcal{A}F)(x) = F_{\mu_0} \circ K^+\Big(F_{\mu_1}^{-1} \circ (K^- F)\Big)(x),$$

If F_α is the fixed point, then

$$r(x, a) = F_\alpha^{-1} \circ F_{\mu_0}(x).$$

Convergence of the iteration $F_\alpha^{(k+1)} = \mathcal{A}F_\alpha^{(k)}$ was analysed for the classical Bass operator in [B. Acciaio et al., 2023].

Experiment

The synthetic market data are generated by a Heston model:

$$dS_t = \sqrt{v_t}\, S_t\, dW_t^{(1)}, \qquad dv_t = \varkappa(\bar{v} - v_t)\, dt + \zeta\sqrt{v_t}\, dW_t^{(2)}, \qquad d\langle W^{(1)}, W^{(2)}\rangle_t = \varrho\, dt,$$

with $\varrho = -0.7$, producing a pronounced negative skew.

We take $t_1 = 0.25, t_2 = 1$ and calibrate three models to the marginals

$$\delta_{S_0} \quad \to \quad \mu_{t_1} \quad \to \quad \mu_{t_2}$$

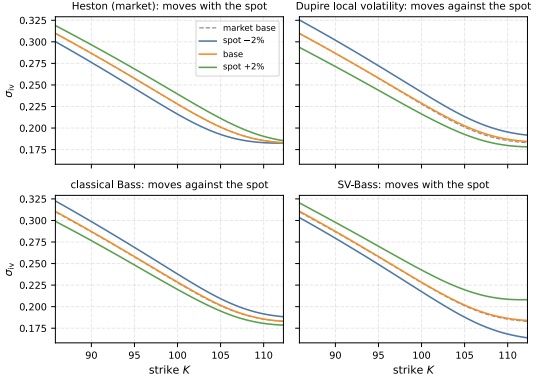
1. the Dupire local volatility model.
2. the classical Bass model.
3. the SV-Bass model with the arithmetic SABR reference:

$$dY_t = \xi_t\, dB_t, \qquad d\xi_t = \gamma\, \xi_t\, dW_t, \qquad d\langle B, W\rangle_t = \rho\, dt,$$

with (ξ_0, ρ, γ) matched to the ATM level and skew of the market.

Smile dynamics

Each calibrated model is held fixed and only its initial state is shifted by $\pm 2\%$ of S_0 - the spot for Heston and Dupire, the reference driver for the two Bass models - and the t_1 smile is recomputed.

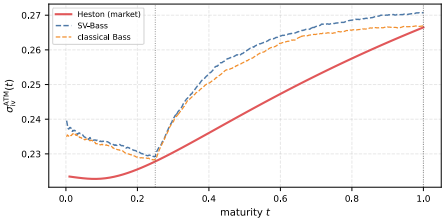


The SV-Bass smile moves *with* the spot and closely tracks Heston.

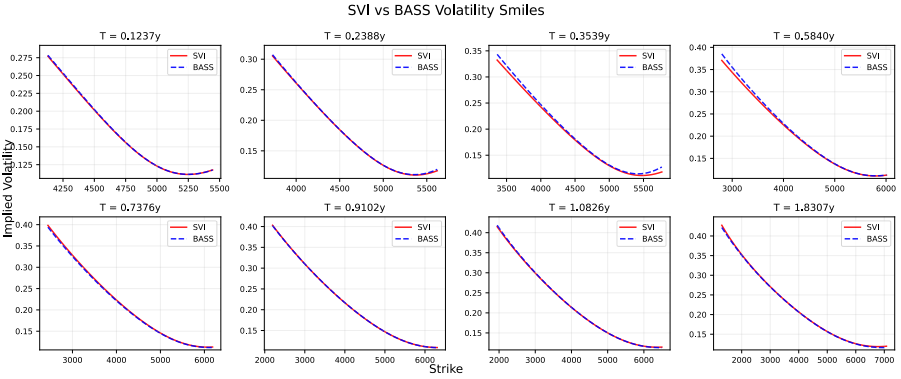
Experiment

The experiments also reveal the limits of the construction:

1. The dynamics of the smiles survive only if the ATM skew of the reference is steep enough.
2. Each restart of the volatility loses part of the spot–vol co-movement.
3. Between the dates the ATM curve follows the classical Bass model.



Calibration to the S&P 500 market (SPX, 21.02.2024)



Transported SABR calibration. Reference: $\xi_0 = 706.6, \quad \rho = -0.706, \quad \gamma = 1.048.$

Summary

1. A new cost functional. The Brownian reference of the Bass construction is replaced by a stochastic volatility one:

$$\mathrm{E}\big\langle X - B \big\rangle_1 \quad \longrightarrow \quad \mathrm{E}\big\langle X - Y \big\rangle_1 .$$

2. Construction of the solution. The optimizer keeps the Bass form, with the reference increment in place of the Gaussian one:

$$X_t = \mathrm{E}\big[g(\tilde{Y}_1) \mid \mathcal{F}_t\big], \qquad \tilde{Y}_1 = r(X_0, \xi_0) + \int_0^1 \xi_s \, dB_s,$$

where g and r come from a one-dimensional fixed-point equation.

3. Further research. The volatility is restarted at every calibration date. This restart is what makes the dynamics fade from one date to the next and the ATM curve kink and bulge in between - the goal is a construction that avoids it.

Key References


J. Backhoff-Veraguas, M. Beiglböck, M. Huesmann, S. Källblad.
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A fundamental theorem for weak martingale optimal transport.
Ann. Appl. Probab., 35(2), 2025.

Appendix

Appendix: Convex order and Strassen's theorem

Definition (Convex order) Let $\mu_0, \mu_1 \in \mathcal{P}_1(\mathbb{R})$. We write $\mu_0 \preceq_{cx} \mu_1$ if

$$\int_{\mathbb{R}} \varphi(x)\, \mu_0(dx) \leq \int_{\mathbb{R}} \varphi(x)\, \mu_1(dx) \quad \text{for every convex function} \quad \varphi : \mathbb{R} \to \mathbb{R}.$$

Theorem. (Strassen) Let $\mu_0, \mu_1 \in \mathcal{P}_1(\mathbb{R})$. There exists a martingale coupling $\pi \in \mathrm{Cpl}(\mu_0, \mu_1)$, that is, a probability measure π on $\mathbb{R} \times \mathbb{R}$ with marginals μ_0 and μ_1 such that

$$\int_{\mathbb{R}} y\, \pi_x(dy) = x \qquad \text{for } \mu_0\text{-a.e. } x,$$

where $\pi(dx, dy) = \mu_0(dx)\pi_x(dy)$, if and only if $\mu_0 \preceq_{cx} \mu_1$.

Appendix: Dupire formula and LSV

Dupire formula.

$$\sigma^2_{\text{loc}}(T,K) = \frac{\partial_T C + rK\,\partial_K C}{\frac{1}{2}K^2\,\partial_{KK} C}.$$

Local Stochastic Volatility. To obtain both the perfect fit and suitable dynamics, a non-parametric function $\sigma(t,S_t)$ is added on top of the stochastic volatility process (v_t):

$$\begin{aligned} dS_t &= rS_t\,dt + \sigma(t,S_t)\,\psi(v_t)\,S_t\,dW_t^{(1)}, \\ dv_t &= a(v_t)\,dt + b(v_t)\,dW_t^{(2)}. \end{aligned}$$

The function $\sigma(t,S_t)$ is again given by a Dupire-type formula:

$$\sigma(t,S_t) = \sqrt{\frac{\sigma^2_{\text{loc}}(t,S_t)}{\mathrm{E}\left[\psi^2(v_t)\mid S_t\right]}}.$$

Appendix: Theorem 1 (Equivalence)

Theorem 1. The dynamic and static problems are equivalent:

$$\sup_{X\in\mathcal{X}(\mu_0,\mu_1;\kappa)} \mathrm{E}\big\langle X,Y\big\rangle_1 = \sup_{\lambda\in \mathrm{Cpl}_{M_1}(\kappa,\mu_1)} \iint \sup_{q\in\mathrm{Cpl}(\lambda_{x,a},\,\theta_a)} \int yz\; q(dy,dz)\; \kappa(dx,da).$$

From the static optimizer λ^* the optimal process X^* can be constructed.

($\leq$) Any admissible X yields $\lambda = \mathrm{Law}(X_0,\xi_0,X_1) \in \mathrm{Cpl}_{M_1}(\kappa,\mu_1)$.

$\mathrm{Law}(X_1,\, Y_1 \mid X_0 = x, \xi_0 = a) \in \mathrm{Cpl}(\lambda_{x,a},\, \theta_a)$ giving a feasible static value.

($\geq$) From a static optimizer λ^* with optimal kernels $q_{x,a} \in \mathrm{Cpl}(\lambda_{x,a},\theta_a)$, let $f_{x,a}$ be the transport map with $(f_{x,a})_{\#}\,\theta_a = \lambda_{x,a}$ (in one dimension it is unique and given by the monotone rearrangement $f_{x,a} = F^{-1}_{\lambda_{x,a}} \circ F_{\theta_a}$), and define

$$X_t = \mathrm{E}\big[f_{X_0,\xi_0}(Y_1) \mid \mathcal{F}_t\big].$$

Then X is admissible and attains the same value.

Appendix: Theorem 2 (Existence of a Static Optimizer)

Assume that $b, \eta : \mathbb{R} \to \mathbb{R}$ are globally Lipschitz and have at most linear growth. For each $a \in \mathbb{R}$, let ξ^a be the unique strong solution of

$$d\xi_t = b(\xi_t)\, dt + \eta(\xi_t)\, dW_t, \qquad \xi_0 = a, \qquad t \in [0, 1].$$

Define

$$Z^a = \int_0^1 \xi_t^a \, dB_t, \qquad \theta_a = \mathrm{Law}(Z^a).$$

Theorem 2. Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ with $\mu_0 \preceq_{cx} \mu_1$ and $\kappa \in \mathrm{Cpl}(\mu_0, \beta)$.

$$\inf_{\lambda \in \mathrm{Cpl}_{M_1}(\kappa, \mu_1)} \iint \mathcal{W}_2^2(\lambda_{x,a}, \theta_a)\, \kappa(dx, da)$$

admits an optimizer $\lambda^* \in \mathrm{Cpl}_{M_1}(\kappa, \mu_1)$.

Appendix: Theorem 2

Proof

Theorem. [Beiglböck et al., 2023] Assume that C is bounded from below, lower semicontinuous on $\mathbb{R} \times \mathcal{P}_p(\mathbb{R})$, and convex in the second variable. Take $\pi_n \in \mathrm{Cpl}(\mu_0, \mu_1)$. If $\pi_n \to \pi$ in $\mathcal{P}_p(\mathbb{R}^2)$ then:

$$\int_{\mathbb{R}} C(x, \pi_x)\, \mu_0(dx) \leq \liminf_{n\to\infty} \int_{\mathbb{R}} C(x, \pi_{n,x})\, \mu_0(dx).$$

(1) The cost $C((x,a),\rho) = \mathcal{W}_2^2(\rho, \theta_a)$ is non-negative, lsc, and convex in ρ. Hence the functional

$$\lambda \mapsto \iint \mathcal{W}_2^2(\lambda_{x,a}, \theta_a)\, \kappa(dx, da) \quad \text{is lsc in } \mathcal{P}_2(\mathbb{R}^3).$$

(2) Compactness of $\mathrm{Cpl}_{M_1}(\kappa, \mu_1)$ in $\mathcal{P}_2(\mathbb{R}^3)$ is proved by compactness of $\mathrm{Cpl}(\kappa, \mu_1)$ together with the martingale constraint being closed under weak convergence.

$$\lambda_n \to \lambda \quad \text{and} \quad \int \varphi(x,a)(y-x)\, d\lambda^n = 0 \quad \forall \varphi \in C_b(\mathbb{R}^2)$$

passing to the limit gives $\int y\, \lambda_{x,a}(dy) = x$, κ-a.s.

Appendix: Theorem 3 (Bass Verification Theorem)

Theorem 3 Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$, and let $\kappa \in \mathrm{Cpl}(\mu_0, \beta)$. Define the reference process:

$$\tilde{Y}_t = r(X_0, \xi_0) + \int_0^t \xi_s \, dB_s, \qquad 0 \le t \le 1.$$

$$\text{Set} \quad \nu = \mathrm{Law}(\tilde{Y}_1), \qquad g = F_{\mu_1}^{-1} \circ F_\nu.$$

Suppose that the function r is chosen so that

$$\mathrm{E}\left[g\left(r(x,a) + \int_0^1 \xi_s^a \, dB_s \right) \right] = x, \qquad \kappa\text{-a.e. } (x,a). \tag{1}$$

Then

$$X_t = \mathrm{E}[g(\tilde{Y}_1) \mid \mathcal{F}_t], \qquad 0 \le t \le 1.$$

is an optimizer of problem

$$\inf_{X \in \mathcal{X}(\mu_0, \mu_1; \kappa)} \mathrm{E}\big\langle X - Y \big\rangle_1.$$

Appendix: Theorem 3

Proof idea

(1) Bound the static problem using the convex conjugate functions ψ and ψ^*:

$$\sup_{\lambda \in \mathrm{Cpl}_{M_1}(\kappa,\mu_1)} \iint \sup_{q \in \mathrm{Cpl}(\lambda_{x,a},\, \theta_a)} \int yz \; q(dy,dz) \; \kappa(dx,da)$$
$$\leq \int \psi^*(y)\, \mu_1(dy) + \iint \psi(z)\, \theta_a^{r(x,a)}(dz)\, \kappa(dx,da) - \int x\, r(x,a)\, \kappa(dx,da),$$

where $\theta_a^{r(x,a)} = (r(x,a) + \mathrm{id})_{\#}\, \theta_a$.

(2) Show attainment of the bound. Since $g = F_{\mu_1}^{-1} \circ F_\nu$ is increasing, write $g = \nabla\psi$ with ψ^* its convex conjugate. Set

$$q_{x,a} = \mathrm{Law}(X_1,\, Y_1 \mid X_0 = x,\, \xi_0 = a) = \mathrm{Law}(g(r(x,a) + Z^a)\,,\, r(x,a) + Z^a)\,.$$

This coupling is supported on the graph $y = \nabla\psi(z)$, so Fenchel equality $yz = \psi(z) + \psi^*(y)$ holds $q_{x,a}$-a.s., and the bound is attained. □

Appendix: Multi-marginal construction

Given marginals $\mu_{t_0}, \mu_{t_1}, \ldots, \mu_{t_n}$, construct the process interval by interval.

1. For each interval $[t_i, t_{i+1}]$, choose one restarting volatility state a_i, for example: $a_i = \mathrm{E}[\xi_{t_i}]$. Compute the reference increment law

$$Z_i = \int_{t_i}^{t_{i+1}} \xi_s^{a_i}\, dB_s, \;\; \theta_i = \mathrm{Law}(Z_i).$$

2. Solve the fixed-point problem: $F_{\alpha_i} = \mathcal{A}_i F_{\alpha_i}$ and store the functions:

$$g_i = F^{-1}_{\mu_{t_{i+1}}} \circ F_{\nu_i}, \qquad r_i(x) = F^{-1}_{\alpha_i} \circ F_{\mu_{t_i}}(x).$$

3. Simulate the process S_t on $[0, t_1]$; on each subsequent interval, given the value $S^-_{t_i}$ at the end of the previous interval, set for $t \in [t_i, t_{i+1}]$:

$$S_t = \mathrm{E}\left[g_i \left(r_i(S^-_{t_i}) + \int_{t_i}^{t_{i+1}} \xi_s^{a_i}\, dB_s \right) \,\middle|\, \mathcal{F}_t \right].$$

Appendix: Choice of the reference parameters

Heston market parameters:

S_0	$\varkappa$	$\bar{v}$	ζ	v_0	ϱ
100	2	0.07	1	0.1	-0.7

The transport reproduces the marginals for *any* reference, so (ξ_0, ρ, γ) are chosen for the smile dynamics they induce.

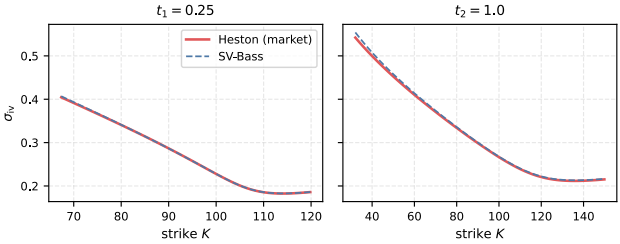
1. The scale is a least-squares Hagan fit to the two market smiles: $\xi_0 = 24.32 \approx s_0 \sigma_{\mathrm{iv}}$.
2. The pair (ρ, γ) is pinned by matching the at-the-money skews of the reference and the market:

$$\left.\frac{\partial \sigma_{\mathrm{ref}}}{\partial K}\right|_{K=s_0} = \left.\frac{\partial \sigma_{\mathrm{mkt}}}{\partial K}\right|_{K=s_0} \quad \Longrightarrow \quad \rho\gamma = -0.93.$$

	ξ_0	ρ	γ
Reference model	24.32	-0.99	0.95

The dynamic parameters of the reference are an *input* of the model: vanilla prices at a single date do not identify the dynamics.

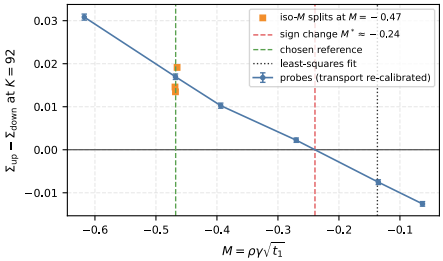
Appendix: Marginal fit



The SV-Bass model reproduces the Heston smiles at t_1, t_2.

Appendix: When the dynamics survive

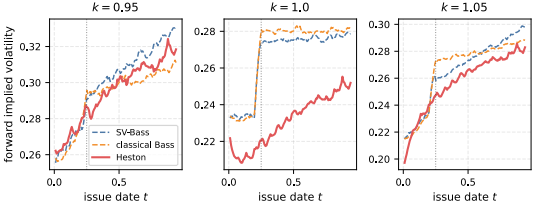
The smile response of the calibrated model is governed by the dimensionless group $M = \rho\gamma\sqrt{t_1}$ of the reference and changes sign at $M^* \approx -0.24$.



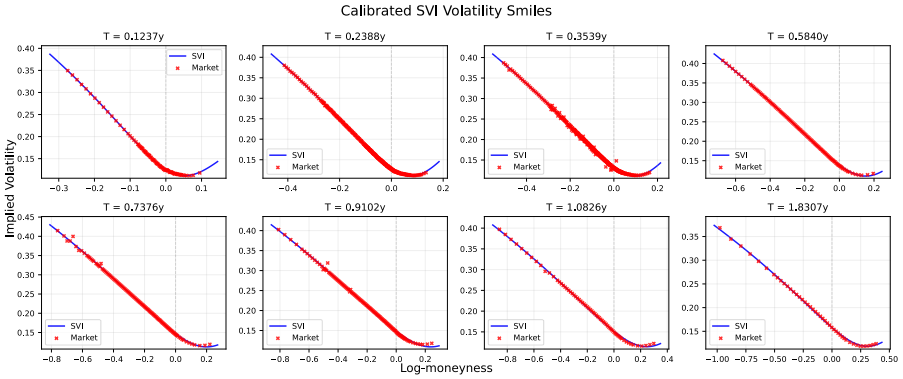
Appendix: Dynamics across calibration dates

Each restart of the volatility at a calibration date loses a fixed amount ($\approx \rho\gamma$ per date) of the spot–vol response: the dynamics decay with the *number of dates*, not with calendar time.

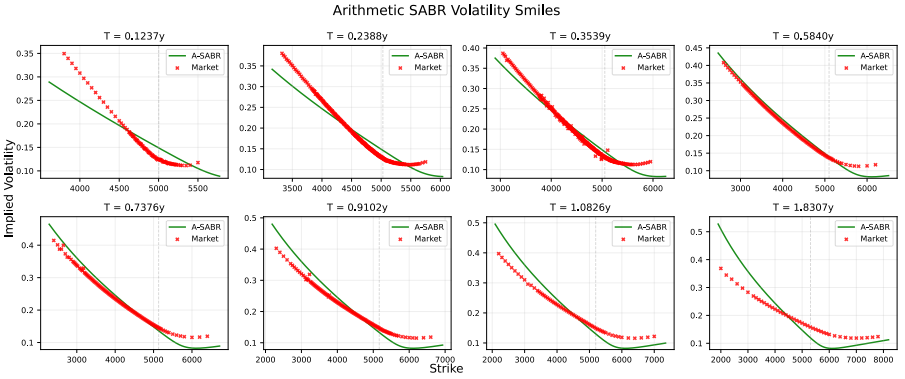
After t_1 the forward volatility of the SV-Bass model tracks the classical Bass model rather than Heston.



Appendix: SPX market data and reference fit



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